SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0612 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Pujitha

Annexure A

Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

Example	Speed	Accuracy	Fitness Value	Suitable Solution
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

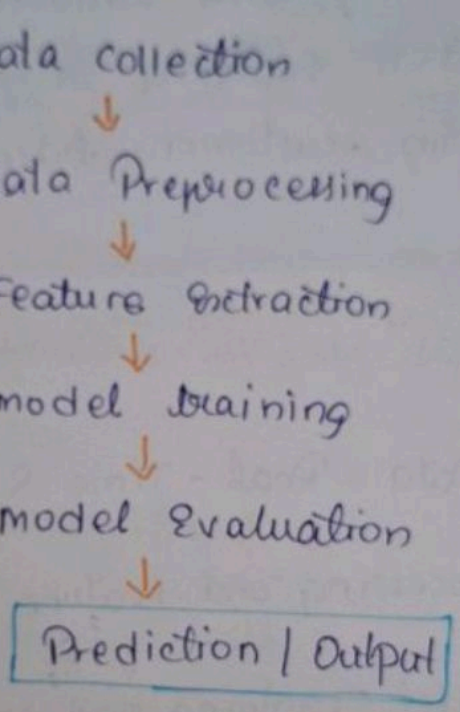
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Functional Blocks of a Machine learning system

A machine learning (ML) system learns Patterns from data and uses them to make predictions or decisions.

Neat sketch



Explanation

1) Data collection : Collect data from databases, sensors.
2) Data Preprocessing : Remove errors, handle missing values, convert data.
3) Feature Extraction: Identify important attributes required for learning, such as age, temperature
4) model training: Train the algorithm using the Prepared data to learn Patterns.
5) model evaluation: Test the model using unseen data and measure its performance.

Supervised vs Unsupervised Learning:

- Supervised learning : Uses labelled data (input with the correct Output).

  Example :- Predicting whether an email spam or not.

- Unsupervised learning: Uses unlabelled data and identifies hidden patterns or groups.

  Example: Grouping customers based on purchasing behaivour.

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2) Machine learning system for Disaster Management

Historical Data + Real-Time Sensor Data.

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Data preprocessing and Feature Selection

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ML Model Training and validation

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Disaster Risk Prediction

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Early warning and Emergency Response

Explanation :-

1) Data collection :- Collect historical rainfall, river levels, seismic activity, soil moisture, weather reports, and real-time sensor data.

2) Preprocessing : Remove duplicate records, handle missing values, normalize numerical values, and

ncode categorical data.

3) Feature selection :- Select important features such as rainfall, river level, Ground movement and soil moisture to improve prediction.

4) Training and Evaluation: Split the historical dataset into Training, Validation and test sets.

3) Back Propagation in a Multilayer Perceptron (MLP)

Back Propagation is a learning algorithm that updates neural network weights by calculating the prediction error

Input layer
↓
Hidden layer
↓
Output layer
↓
Calculate Error
↓
update weights Backward

Working

1) Forward Pass :- Input features are Passed through the hidden layer to produce an output. Assume the actual output is 1.

2) Error Calculation: For Squared Error, the loss is

$$e = \frac{1}{2}(y - \hat{y})^2$$

$$E = \frac{1}{2}(1-0.6)^2 = 0.08$$

3) Backward propgation: The Error Gradients are calculated from the output layer toward the hidden layer.

4) weight update:- The weight are adjusted using gradient descent.

$$w_{new} = w_{old} - \eta \frac{\partial E}{\partial w}$$

Here, η is the learning rate.

5) Repeat: The Process continues for multiple training iterations until the Error becomes small.

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4) Genetic Algorithm for Maximizing Fitness

A Genetic Algorithm (GA) is an optimization technique inspired by natural selection. It improves candidate solutions through selection, crossover & mutation.

Ex	Speed	Accuracy	Fitness	suitable
1	High	Low	40	NO
2	medium	medium	60	Yes
3	High	medium	75	Yes
4	Low	High	65	Yes.
5	medium	High	80	Yes

Step 1:- Initial Population
The five given Examples form the Initial Population Each candidate is Evaluated using its fitness value.

Step 2 :- Selection
Candidates with higher fitness values have a Greater chance of being selected as Parents. Examples 5 & 3, with fitness values 80 & 75, are strong Candidates

Step 3 :- Crossover
Selected Parents Exchange Parts of their Encoded solutions to produce offspring.

Step 4:- mutation:
mutation randomly changes a Parameter to Explore new solution. For Ex speed may change from medium to high, Creating a new Candidate.

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3. Neural Network classification and Performance Evaluation.

1. A Neural Network can classify whether a Patient has a disease using age, Blood Pressure, Glucose level, and lifestyle Risk as input features.

Training process

1) Input : Collect network the four Patient attributes and the actual disease label (Yes / No).

2) Preprocessing :- Handle missing values, Encode High/ Normal / Low into numerical values, and scale numerical features.

3) Forward Pass :- Pass the input features through the hidden layer and sigmoid output layer to predict disease probability.

4) Back Propagation: Calculate gradients and update the weights to reduce the Error. Repeat across training examples.

5) Evaluation: Compare predictions with actual labels on unseen test data.

Confusion matrix calculation:

Treat Yes as the Positive class.

Example	Actual	Predicted	Result
1	Yes	Yes	TP
2	Yes	NO	FN
3	NO	NO	TN
4	Yes	Yes	TP
5	NO	Yes	FP

Evaluation Metrics

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{2+1}{5} \times 100 = 60\%$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{3} \times 100 = 66.67\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{3} \times 100 = 66.67\%$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 66.67\%$$